

SIMATS ENGINEERING

ASSESSMENT TOOL 2 - Medium(Assignment 1)

COURSE CODE: CSA09

COURSE NAME: Programming in Java COURSE

OUTCOME COVERED

CO2: *Implement Collection Framework interfaces such as List, ArrayList, Set, Map, HashTable, and HashMap using iterators and generics to store, retrieve, and process groups of objects in web applications.(BL3,BL4)*

Assessment Tool Weightage: 10%

QUESTIONS: 40 (each student assigned distinct questions) Total Marks: 100

ANNEXURE A

Assignment 1

Code of Conduct:

*I **N.Adi Narayana** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding				
Algorithm				
Code Implementation				
Competitive Performance				
Test Case Handling				

Annexure A

Assignment 1

Question 36 – Cyber Security Log

A company stores security event logs.

Questions

a) Develop a Java program using StringBuilder.

```
import java.util.*;
```

```
public class CyberSecurityLogAnalyzer {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        StringBuilder logs = new StringBuilder();
```

```
        System.out.print("Enter the number of log messages: ");
```

```
        int n = sc.nextInt();
```

```
        sc.nextLine();
```

```
        // Read log messages
```

```
        for (int i = 1; i <= n; i++) {
```

```
            System.out.print("Enter Log " + i + ": ");
```

```
            String log = sc.nextLine();
```

```
            logs.append(log).append("\n");
```

```
        }
```

```

// Display all logs

System.out.println("\nStored Security Logs:");

System.out.println(logs);

}

```

}Implementation:

The screenshot shows an IDE with a file named `CyberSecurityLogAnalyzer.java`. The code on the left is as follows:

```

1 import java.util.*;
2
3 public class CyberSecurityLogAnalyzer {
4     public static void main(String[] args) {
5
6         Scanner sc = new Scanner(System.in);
7         StringBuilder logs = new StringBuilder();
8
9         System.out.print("Enter the number of log messages: ");
10        int n = sc.nextInt();
11        sc.nextLine();
12
13        // Read log messages
14        for (int i = 1; i <= n; i++) {
15            System.out.print("Enter Log " + i + ": ");

```

The output window on the right shows the following interaction:

```

Enter the number of log messages: 4
Enter Log 1: Login Failed
Enter Log 2: Virus Detected
Enter Log 3: Virus Detected
Enter Log 4: Firewall Alert

Stored Security Logs:
Login Failed
Virus Detected
Virus Detected
Firewall Alert

=== Code Execution Successful ===

```

b) Navigate articles using ListIterator.

```
import java.util.*;
```

```
public class RepeatedLogs {
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter the number of log messages: ");
        int n = sc.nextInt();
        sc.nextLine();
```

```
        String[] logs = new String[n];
```

```
        // Input logs
        for (int i = 0; i < n; i++) {
            System.out.print("Enter Log " + (i + 1) + ": ");
            logs[i] = sc.nextLine();
        }
```

```
        // Find repeated logs
        System.out.println("\nRepeated Log Messages:");
```

```
        boolean found = false;
```

```
        for (int i = 0; i < n; i++) {
            int count = 1;
```

```
            if (logs[i].equals("Visited"))
                continue;
```

```

        for (int j = i + 1; j < n; j++) {
            if (logs[i].equals(logs[j])) {
                count++;
                logs[j] = "Visited";
            }
        }

        if (count > 1) {
            System.out.println(logs[i] + " - " + count + " times");
            found = true;
        }
    }

    if (!found) {
        System.out.println("No repeated log messages.");
    }

    sc.close();
}
}

```

```

1 import java.util.*;
2
3 public class RepeatedLogs {
4     public static void main(String[] args) {
5
6         Scanner sc = new Scanner(System.in);
7
8         System.out.print("Enter the number of log messages: ");
9         int n = sc.nextInt();
10        sc.nextLine();
11
12        String[] logs = new String[n];
13
14        // Input logs
15        for (int i = 0; i < n; i++) {

```

```

Enter the number of log messages: 4
Enter Log 1: Login Failed
Enter Log 2: Virus Detected
Enter Log 3: Firewall Alert
Enter Log 4: Login Failed

Repeated Log Messages:
Login Failed - 2 times

=== Code Execution Successful ===

```

c) Compare String and StringBuilder

String

- 1) Immutable (cannot be changed).
- 2) Creates a new object for every change..
- 3) Slower for repeated string operations.
- 4) Uses concat() or + operator.

StringBulider

- 1) Mutable (can be modified).
- 2) Modifies the existing object.
- 3) Faster for repeated append and update operation
- 4) Uses append(), insert(), delete(), etc

c) Explain Log Analysis

Log analysis is the process of examining security event logs to identify suspicious activities, repeated events, errors, and security threats.

Importance of Log Analysis

- Detects unauthorized login attempts.
- Identifies repeated error or warning messages.
- Monitors system activities.
- Helps in troubleshooting system issues.
- Improves cyber security by detecting attacks early.

Example

Log Message	Analysis
Login Failed	Check for unauthorized access attempts.
Virus Detected	Scan and remove malware.
Firewall Alert	Investigate possible network attack.
Login Failed (Repeated)	May indicate a brute-force attack.